

Database System Engineering and Distributed Backend Development

Course Code: 25CS13002E

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Query 1: -CREATE DATABASE

```
CREATE DATABASE IF NOT EXISTS bookflow7_db;  
USE bookflow7_db;
```

Output: -

✓ 1	16:06:18	CREATE DATABASE IF NOT EXISTS bookflow7_db	1 row(s) affected	0.015 sec
✓ 2	16:06:21	USE bookflow7_db	0 row(s) affected	0.000 sec

Query 2: - CREATE TABLE

```
2520030597-Vidhath  
CREATE TABLE bank_transactions (  
  txn_id INT PRIMARY KEY,  
  customer_name VARCHAR(50),  
  branch_name VARCHAR(50),  
  transaction_type VARCHAR(20),  
  amount DECIMAL(10,2),  
  transaction_date DATE  
);
```

Output: -

✓ 3	16:09:19	CREATE TABLE bank_transactions (txn_id...	0 row(s) affected	0.063 sec
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Query 3: - Insert and Display

```

INSERT INTO bank_transactions VALUES
(101, 'Vidhath', 'Hyderabad', 'Deposit', 5000, '2024-01-05'),
(102, 'Raghuveer', 'Hyderabad', 'Withdrawal', 2000, '2024-01-06'),
(103, 'Nihitha', 'Vijayawada', 'Deposit', 12000, '2024-01-08'),
(104, 'Ananya', 'Vizag', 'Deposit', 8000, '2024-01-10'),
(105, 'Rushika', 'Hyderabad', 'Withdrawal', 3500, '2024-01-11'),
(106, 'Priyadarshini', 'Vizag', 'Deposit', 15000, '2024-01-12'),
(107, 'Amulya', 'Vijayawada', 'Withdrawal', 1000, '2024-01-13'),
(108, 'Akshaya', 'Hyderabad', 'Deposit', 9000, '2024-01-14'),
(109, 'Srujana', 'Vizag', 'Withdrawal', 4000, '2024-01-15'),
(110, 'Varshini', 'Vijayawada', 'Deposit', 11000, '2024-01-16');
SELECT * FROM bank_transactions;

```

Output: -

✓ 4	16:12:54	INSERT INTO bank_transactions VALUES (1...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0.000 sec
✓ 5	16:13:06	SELECT * FROM bank_transactions LIMIT 0...	10 row(s) returned 0.000 sec / 0.000 sec

Result Grid		Filter Rows:	Export: Wrap Cell Content			
	txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
▶	101	Vidhath	Hyderabad	Deposit	5000.00	2024-01-05
	102	Raghuveer	Hyderabad	Withdrawal	2000.00	2024-01-06
	103	Nihitha	Vijayawada	Deposit	12000.00	2024-01-08
	104	Ananya	Vizag	Deposit	8000.00	2024-01-10
	105	Rushika	Hyderabad	Withdrawal	3500.00	2024-01-11
	106	Priyadarshini	Vizag	Deposit	15000.00	2024-01-12
	107	Amulya	Vijayawada	Withdrawal	1000.00	2024-01-13
	108	Akshaya	Hyderabad	Deposit	9000.00	2024-01-14
	109	Srujana	Vizag	Withdrawal	4000.00	2024-01-15
	110	Varshini	Vijayawada	Deposit	11000.00	2024-01-16

Query 4: - SUM()

```

2520030597-Vidhath
SELECT SUM(amount) AS Total_Amount
FROM bank_transactions;

```

Output: -

✓ 6	16:15:55	SELECT SUM(amount) AS Total_Amount FROM...	1 row(s) returned 0.000 sec / 0.000 sec
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Result Grid	Filter Rows:	Export: Wrap Cell Content
Total_Amount		
▶ 70500.00		

Query 5: - AVG()

```

2520030597-Vidhath
SELECT AVG(amount) AS Average_Transaction
FROM bank_transactions;

```

Output: -

✔ 7	16:18:11	SELECT AVG(amount) AS Average_Transacti...	1 row(s) returned	0.015 sec / 0.000 sec
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Result Grid	Filter Rows:	Export: Wrap Cell Content
Average_Transaction		
▶ 7050.000000		

Query 6: -MAX()

```

2520030597-Vidhath
SELECT MAX(amount) AS Highest_Transaction
FROM bank_transactions;

```

Output: -

✔ 8	16:19:46	SELECT MAX(amount) AS Highest_Transacti...	1 row(s) returned	0.000 sec / 0.000 sec
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Result Grid	Filter Rows:	Export: Wrap Cell Content
Highest_Transaction		
▶ 15000.00		

Query 7:- MIN()

```
2520030597-Vidhath
SELECT MIN(amount) AS Lowest_Transaction
FROM bank_transactions;
```

Output: -

✓ 9 16:21:29 SELECT MIN(amount) AS Lowest_Transaction... 1 row(s) returned 0.000 sec / 0.000 sec

Result Grid	Filter Rows:	Export: Wrap Cell Content
Lowest_Transaction		
▶ 1000.00		

Query 8: -COUNT()

```
2520030597-Vidhath
SELECT COUNT(*) AS Total_Transactions
FROM bank_transactions;
```

Output: -

✓ 10 16:23:22 SELECT COUNT(*) AS Total_Transactions F... 1 row(s) returned 0.000 sec / 0.000 sec

Result Grid	Filter Rows:	Export: Wrap Cell Content
Total_Transactions		
▶ 10		

Query 9: -WHERE + SUM()

```
SELECT SUM(amount) AS Total_Deposit
FROM bank_transactions
WHERE transaction_type='Deposit';
```

Output: -

✓ 11 16:25:08 SELECT SUM(amount) AS Total_Deposit FRO... 1 row(s) returned 0.000 sec / 0.000 sec

Result Grid	Filter Rows:	Export: Wrap Cell Content
▶ Total_Deposit		
▶ 60000.00		

Query 10: - GROUP BY

```
2520030597-Vidhath
SELECT branch_name,
       SUM(amount) AS Total_Amount
FROM bank_transactions
GROUP BY branch_name;
```

Output: -

✓ 12 16:26:59	SELECT branch_name, SUM(amount) AS Tota...	3 row(s) returned	0.000 sec / 0.000 sec
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Result Grid	Filter Rows:	Export: Wrap Cell Content
▶ branch_name	Total_Amount	
▶ Hyderabad	19500.00	
Vijayawada	24000.00	
Vizag	27000.00	

Query 11: - GROUP BY + HAVING

```
2520030597-Vidhath
SELECT branch_name,
       SUM(amount) AS Total_Amount
FROM bank_transactions
GROUP BY branch_name
HAVING SUM(amount) > 20000;
```

Output: -

✓ 13 16:28:47	SELECT branch_name, SUM(amount) AS Tota...	2 row(s) returned	0.000 sec / 0.000 sec
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Result Grid	Filter Rows:	Export: Wrap Cell Content
▶ branch_name	Total_Amount	
▶ Vijayawada	24000.00	
Vizag	27000.00	

Query 12: - GROUP BY + ORDER BY

```
2520030597-Vidhath
SELECT branch_name,
       SUM(amount) AS Total_Amount
FROM bank_transactions
GROUP BY branch_name
ORDER BY Total_Amount DESC;
```

Output: -

✓ 14 16:30:48 SELECT branch_name, SUM(amount) AS Total_Amount 3 row(s) returned 0.016 sec / 0.000 sec

Result Grid		Filter Rows:	Export: Wrap Cell Content
	branch_name	Total_Amount	
▶	Vizag	27000.00	
	Vijayawada	24000.00	
	Hyderabad	19500.00	

Query 13: - GROUP BY + HAVING + ORDER BY

```
2520030597-Vidhath
SELECT branch_name,
       COUNT(*) AS Total_Transactions
FROM bank_transactions
GROUP BY branch_name
HAVING COUNT(*) >= 3
ORDER BY Total_Transactions DESC;
```

Output: -

✓ 15 16:32:30 SELECT branch_name, COUNT(*) AS Total_Transactions 3 row(s) returned 0.016 sec / 0.000 sec

Result Grid		Filter Rows:	Export: Wrap Cell Content
	branch_name	Total_Transactions	
▶	Hyderabad	4	
	Vijayawada	3	
	Vizag	3	

Query 14: - WHERE + COUNT()

```
SELECT COUNT(*) AS Withdrawals
FROM bank_transactions
WHERE transaction_type='Withdrawal';
```

Output: -

✓ 16 16:34:22 SELECT COUNT(*) AS Withdrawals FROM ban... 1 row(s) returned 0.000 sec / 0.000 sec

Result Grid	Filter Rows:	Export: Wrap Cell Content
Withdrawals		
▶ 4		

Query 15: - WHERE + AVG()

```
SELECT AVG(amount) AS Avg_Deposit
FROM bank_transactions
WHERE transaction_type='Deposit';
```

Output: -

✓ 18 16:36:57 SELECT AVG(amount) AS Avg_Deposit FROM ... 1 row(s) returned 0.000 sec / 0.000 sec

Result Grid	Filter Rows:	Export: Wrap Cell Content
Avg_Deposit		
▶ 10000.000000		

Query 16: - GROUP BY + MAX()

```
2520030597-Vidhath
SELECT branch_name,
       MAX(amount) AS Highest_Amount
FROM bank_transactions
GROUP BY branch_name;
```

Output: -

✓ 19 16:39:15 SELECT branch_name, MAX(amount) AS High... 3 row(s) returned 0.000 sec / 0.000 sec

Result Grid		Filter Rows:	Export: Wrap Cell Content
	branch_name	Highest_Amount	
►	Hyderabad	9000.00	
	Vijayawada	12000.00	
	Vizag	15000.00	

Query 17: - GROUP BY + MIN()

```
2520030597-Vidhath
SELECT branch_name,
       MIN(amount) AS Lowest_Amount
FROM bank_transactions
GROUP BY branch_name;
```

Output: -

✓ 20 16:40:59	SELECT branch_name, MIN(amount) AS Lowe...	3 row(s) returned	0.000 sec / 0.000 sec
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Result Grid		Filter Rows:	Export: Wrap Cell Content
	branch_name	Lowest_Amount	
►	Hyderabad	2000.00	
	Vijayawada	1000.00	
	Vizag	4000.00	

Query 18: - WHERE + ORDER BY

```
2520030597-Vidhath
SELECT *
FROM bank_transactions
WHERE amount > 8000
ORDER BY amount DESC;
```

Output: -

✓ 21 16:42:50	SELECT * FROM bank_transactions WHERE a...	4 row(s) returned	0.000 sec / 0.000 sec
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Result Grid		Filter Rows:	Export: Wrap Cell Content			
	txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
▶	106	Priyadarshini	Vizag	Deposit	15000.00	2024-01-12
	103	Nihitha	Vijayawada	Deposit	12000.00	2024-01-08
	110	Varshini	Vijayawada	Deposit	11000.00	2024-01-16
	108	Akshaya	Hyderabad	Deposit	9000.00	2024-01-14

Query 19: - GROUP BY + COUNT()

```
2520030597-Vidhath
SELECT branch_name,
       COUNT(*) AS Customer_Count
FROM bank_transactions
GROUP BY branch_name;
```

Output: -

✓ 22 16:44:47 SELECT branch_name, COUNT(*) AS Custome... 3 row(s) returned 0.000 sec / 0.000 sec

Result Grid		Filter Rows:	Export: Wrap Cell Content
	branch_name	Customer_Count	
▶	Hyderabad	4	
	Vijayawada	3	
	Vizag	3	

Query 20: - HAVING + COUNT()

```
2520030597-Vidhath
SELECT branch_name,
       COUNT(*) AS Total_Transactions
FROM bank_transactions
GROUP BY branch_name
HAVING COUNT(*) > 3;
```

Output: -

✓ 23 16:51:28 SELECT branch_name, COUNT(*) AS Total_T... 1 row(s) returned 0.000 sec / 0.000 sec

Result Grid		Filter Rows:	Export: Wrap Cell Content
	branch_name	Total_Transactions	
▶	Hyderabad	4	

Query 21: - GROUP BY + AVG()

```
2520030597-Vidhath
SELECT branch_name,
       AVG(amount) AS Avg_Amount
FROM bank_transactions
GROUP BY branch_name;
```

Output: -

✓ 24 16:53:14 SELECT branch_name, AVG(amount) AS Avg_... 3 row(s) returned 0.000 sec / 0.000 sec

Result Grid		Filter Rows:	Export: Wrap Cell Content
	branch_name	Avg_Amount	
▶	Hyderabad	4875.000000	
	Vijayawada	8000.000000	
	Vizag	9000.000000	

Query 22: - WHERE + GROUP BY + HAVING

```
2520030597-Vidhath
SELECT branch_name,
       SUM(amount) AS Total_Deposit
FROM bank_transactions
WHERE transaction_type='Deposit'
GROUP BY branch_name
HAVING SUM(amount) > 15000;
```

Output: -

✓ 25 16:54:58 SELECT branch_name, SUM(amount) AS Total... 2 row(s) returned 0.000 sec / 0.000 sec

Result Grid		Filter Rows:	Export: Wrap Cell Content
	branch_name	Total_Deposit	
▶	Vijayawada	23000.00	
	Vizag	23000.00	

Query 23: - GROUP BY + HAVING + ORDER BY

```
2520030597-Vidhath
SELECT branch_name,
       AVG(amount) AS Avg_Amount
FROM bank_transactions
GROUP BY branch_name
HAVING AVG(amount) > 7000
ORDER BY Avg_Amount DESC;
```

Output: -

✓ 26 16:56:27 SELECT branch_name, AVG(amount) AS Avg_... 2 row(s) returned 0.000 sec / 0.000 sec

Result Grid		Filter Rows:	Export: Wrap Cell Content
	branch_name	Avg_Amount	
▶	Vizag	9000.000000	
	Vijayawada	8000.000000	